

Circuit World: A Multiplayer VE for Researching Engineering Learning

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ABSTRACT

Business travel leads to the consumption of large amounts of energy. Circuit World (CW), a Unity virtual environment, uses avatars and realistic simulations of electrical circuits to test the effectiveness of teaching complex engineering tasks without travel. To evaluate CW, a study compares how well participants learn to repair a circuit when trained face-to-face, via Zoom, and within CW wearing an HMD. After training and completion of a knowledge quiz, learners are given a circuit to repair by themselves using a non-immersive desktop version of CW. Participants repair it again two weeks later to test retention. Each study session involves three people: learner, trainer, and researcher. The learner and trainer control avatars and the researcher, invisible, observes over the learner's shoulder. Collected data includes task performance logs from CW, knowledge quiz scores, and, based on video recordings, eye gaze

tracking, behavioral coding, and measures of facial synchrony between trainer and learner.

Keywords: virtual reality, training, circuits, avatar, remote collaboration, virtual environment

Index Terms: [Human-centered computing: Human computer interaction (HCI): Interaction paradigms]: Virtual reality; [Human-centered computing: Human computer interaction (HCI)]: Empirical studies in HCI

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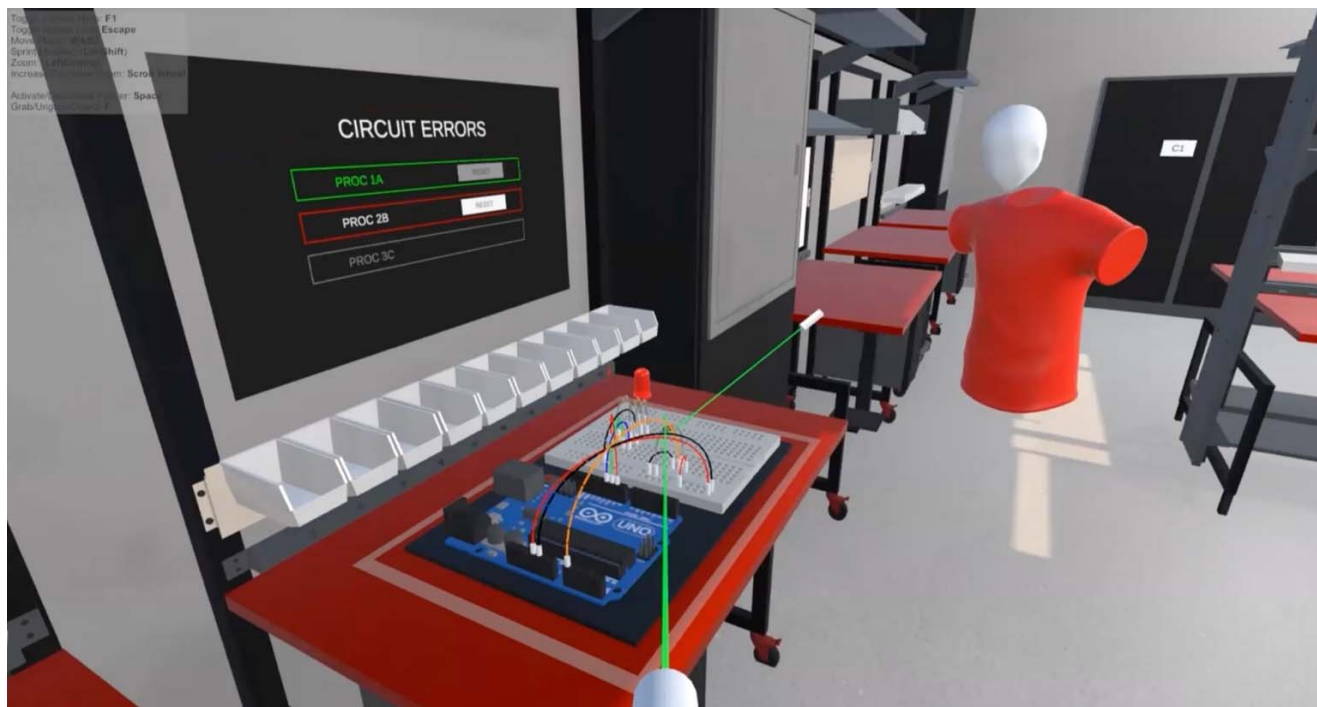


Figure 1. Circuit World is a multiplayer virtual environment that allows collaborative assembly of electrical circuits that function similarly to real-world circuits, which is useful for training and circuit design when real-world hardware is not available.